

16.7 A 40-310TOPS/W SRAM-Based All-Digital Up to 4b In-Memory Computing Multi-Tiled NN Accelerator in FD-SOI 18nm for Deep-Learning Edge Applications

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In-memory computing (IMC) has been proposed to address compute-intensive data-driven AI workloads, using either SRAM or emerging memory technologies such as PCM, RRAM, and MRAM offering different trade-offs when used as an integrated computing device at the system level. A notable distinction is between digital vs. analog IMC. The latter uses either resistive or capacitive sharing techniques to maximize row parallelism, but at the expense of inaccuracies and accumulation resolution loss due to device variations across PVT and the limited SNR and dynamic range of the ADC/readout circuits. Most of the analog SRAM IMC solutions make use of large logic bit cells and aggressive ADC/readout bitwidth reduction leading to low memory density and computing inaccuracy. These drawbacks significantly limit deployment when functional safety, low-cost testing and system scalability to handle general-purpose workloads are required. In contrast, the deterministic behavior of digital IMC, and compatibility with pushed technology scaling rules offer a fast path for the next generation of neural processing systems. However, the integration of IMC into a Neural Processing Unit (NPU) must retain a mix of computing capabilities, while aiming at a substantial improvement in terms of power and cost efficiency. In this work, we present the architecture of a scalable and design time parametric NPU for edge AI relying on digital SRAM IMC (DIMC) using 8T standard bitcells integrated into IMC tiles supporting 1, 2, and 4b operation (a version with 8b support is in the works), instantiated in multiple clusters with digital logic and driven by a custom tensor slicing optimizing graph compiler, achieving an end-to-end system-level energy efficiency from 40-310TOPS/W in 18nm FD-SOI.

Figure 16.7.1 depicts the top level view including an ARM Cortex M host CPU, a tightly coupled memory (TCM), JTAG, I/Os, clock and reset factory, system memory, and 8 NPU clusters each with four tensor streaming DMAs, two 64b AXI4 ports, 8 IMC computing tiles of 32Kb. Tiles rely on a pushed rule 8T bitcell design and can be configured to either function as a computing tile or as a standard memory offering 1R1W functionality via the AXI bus, with a marginal incremental area cost. This provides an advantage when replacing a portion of the on-chip system memory, while dynamically trading off compute vs storage needs.

The NPU has a dataflow architecture relying on 64b streaming links, tensor DMAs and DIMC tiles integrated into a cluster with control logic. Dataflow and multicasting are controlled via a back-pressure mechanism, while data filtering and context switching simplify NN graph mapping. The DMAs support a large variety of access patterns and data manipulation at bit granularity to efficiently read/write data and support kernel-sized sub-windows traversal to linearize activation data. All subunits are accessible via an on-chip interconnect and are controlled via configuration registers. Through SW, reconfigurable chains can be created connecting DMAs, IMC tiles, and other units with stream link ports to consume entire (fused) network layers or subsets with a single configuration step; a dedicated HW unit (Epoch Controller) enables full independent processing, an interrupt controller signals relevant events to the host CPU.

Up to 8 DIMC tiles can be grouped (Fig. 16.7.2) or used independently, the cluster logic allows activation data broadcasting to multiple IMCs to process kernels in parallel, as well as their chaining for larger kernels. A mix of both approaches together with striping can be used to optimize a given network layer. The weight data is loaded into the tiles before the inference starts if it fits entirely or is loaded on demand (Fig. 16.7.4). An AXI slave port is used to access data when tiles are configured as memory, while dedicated interfaces stream activation data to each tile through a 1024b feature buffer and automatically trigger execution. The multicast support and bit field selectors of each input interface can be used to forward partial sums packed as 4 16b values. Each PSOUT interface and an output mux [OUT MUX] enable the interleaved forwarding of output data to a single data stream, while the PACK MUX interface can be used to pack the output streams into 4 16b data values. All output ports support bit shifting, rounding and sign-extend operations. All tiles include an activation unit supporting RELU operation when needed; final results are sent to the tensor DMAs to be stored back in memory.

Figure 16.7.3 shows the architecture of a 32Kb DIMC tile using a bitcell with pushed rule 8T 1R1W arranged in multiple core segments that behave as a single array in memory-mapped mode or are configured in multiple independent sub-arrays during read access IMC mode. The core structure supports multiple independent read bitlines (RBL) for each subarray, shown as RBL0/1/2/3..P-1. In memory mode, a single RBL in each column is accessed through one of the read word lines (RWL) and is sensed and latched to Q<N>, while in compute mode multiple RBLs are accessed in each column through multiple RWLs and sensed in parallel as separate paths in IO blocks and sent to the interleaved computational slices. This flexibility allows a design-time parametric amount of parallel multibit MACs to be associated per Kb of kernel storage. In this work, a single DIMC instance has a total of 256 parallel signed/unsigned 4b MAC operations/cycle split into 4 subarrays of 64 4b MAC that share a common accumulation pipeline. Parallel MAC arrays are also runtime configurable as 512 2b and 1024 1b MACs per cycle.

The timing diagram and the IO circuit that highlight both modes are shown in Fig. 16.7.3. The recombination data slice supports a partial sum input (PSIN) for cascading multiple tiles with a final accumulation 16b output. The decoder circuit supports the generation of multiple RWLs in computing mode, while maintaining the unified RWL selection in memory mode. The write circuit uses a mux factor of 1 providing a large kernel write bandwidth of 256b/cycle for each DIMC tile. The tiles share a single supply and body bias domain for the bitcells and MAC arrays, and the periphery circuitry. Programmable write- and read-assist techniques based on negative bitline and wordline lowering are used to enable low-voltage performance down to 0.525V across the industrial PVT range with a forward body bias (FBB) up to 1.5V.

Figure 16.7.4 shows the mapping for a VGG16-style network reduced to fit in the on-chip memory, as well as the configurations chosen for a single-cluster inference (measured) and all 8 clusters (estimated), a compiler slices layers performing tensor striping, chaining and parallel execution on the DIMC tiles to optimize tile usage efficiency based on layers' topologies. If all 8 clusters are used, all layers' kernels fit into the DIMC storage, greatly reducing the throughput for activations reloads. On this benchmark, the NPU executes 693 and 4776inf/sec with 1 and 8 clusters, respectively, for a total of 1.25GMACs/inf.

Figure 16.7.7 shows the die photo of the chip that measures 30.9mm² in 18nm FD-SOI technology. The 8 IMNPU subsystem measures 4.2mm² (3300x1275μm²) with a peak throughput of 57TOPS for 4b weights and activation and includes 2Mb of capacity made up of a total of 64 DIMC tiles. The rest of the chip area is consumed by the CPU, interconnect, system DMAs, I/Os, system memory, analog, digital and connectivity IPs. An example of an always-on battery-operated video surveillance application is also shown using a staged inference operating mode.

Figure 16.7.5 shows the silicon measurement of energy efficiency in the 4b and 1b modes measured at 0.525V across a range of kernel data sparsity and inter-kernel transition density. The graphs show the single-tile energy efficiency and that of the full NPU cluster for the same conditions, as well as the diminishing returns in terms of the TOPS/W gains for the single DIMC instance not translating proportionally at the NPU cluster level due to scalar processing and data movement overheads. Fig. 16.7.5 also shows compute density gain expressed in TOPS/mm² across the range of FBB and its correlation to the energy efficiency TOPS/W at a fixed supply of 0.525V with a peak of 1.9x for 1.5V FBB with only a 14% reduction in energy efficiency. The measured shmoo of the DIMC tile operational frequency is also shown with the TOPS/W and TOPS/mm² at different supply voltage values. The peak energy efficiency vs. precision for weights and activations was 77TOPS/W for the 4b mode and 310TOPS/W for the 1b mode with a peak density of 13.6TOPS/mm² for 4b and 54TOPS/mm² for 1b. In summary, this work presents a scalable DIMC-based NPU accelerator with deterministic output supporting full accumulator resolution computation. The comparison table in Fig. 16.7.6 of FOM normalized to 4b weights/activations shows the comparison with previously published non-IMC and IMC accelerators.

References:

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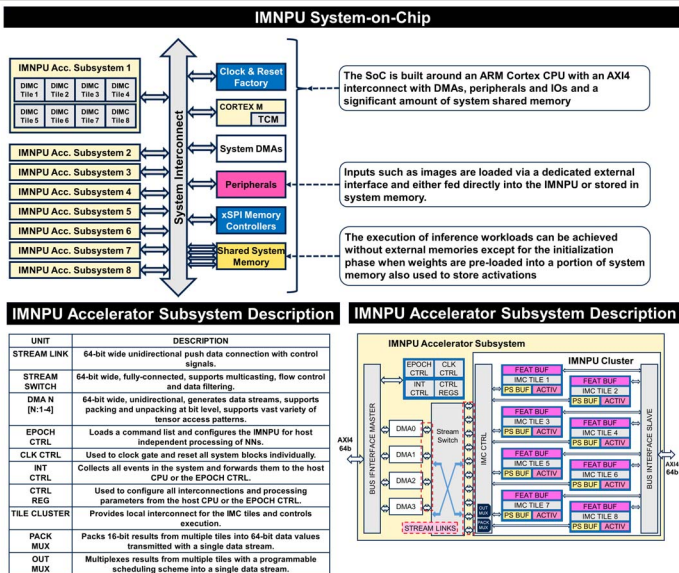


Figure 16.7.1: Full prototype system including multi-cluster In-memory NPU (IMNPU).

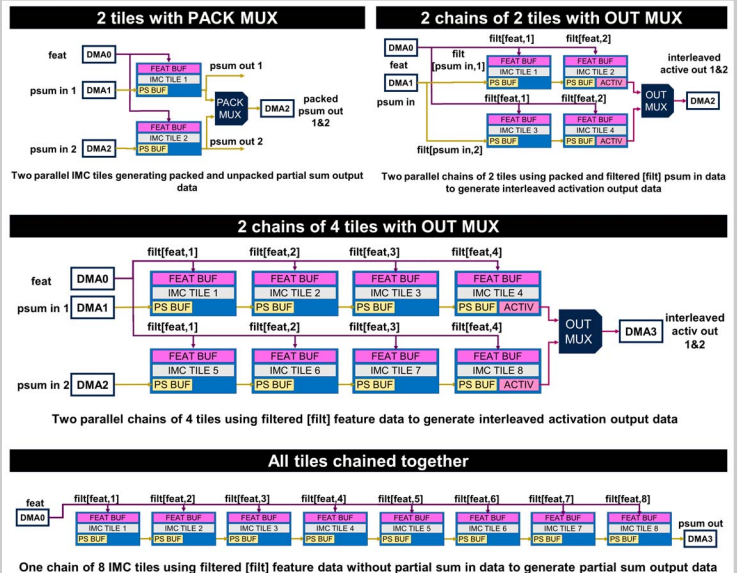


Figure 16.7.2: IMNPU chaining configuration.

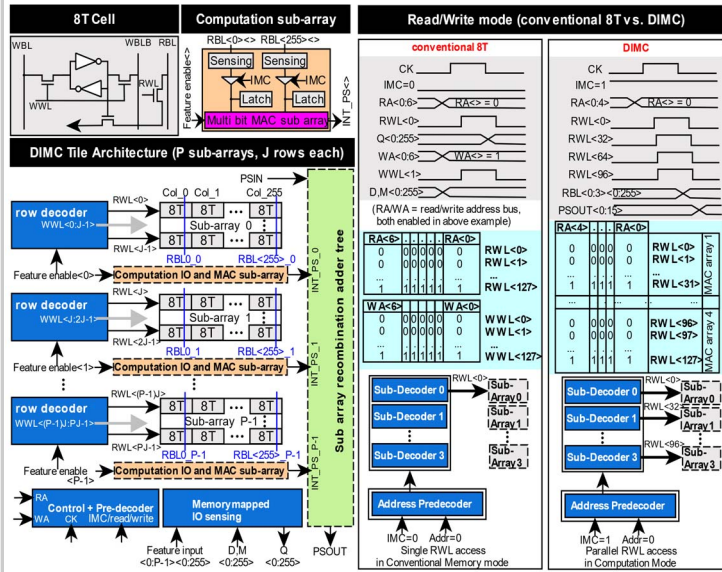


Figure 16.7.3: DIMC single tile description with duality of memory and computational modes.

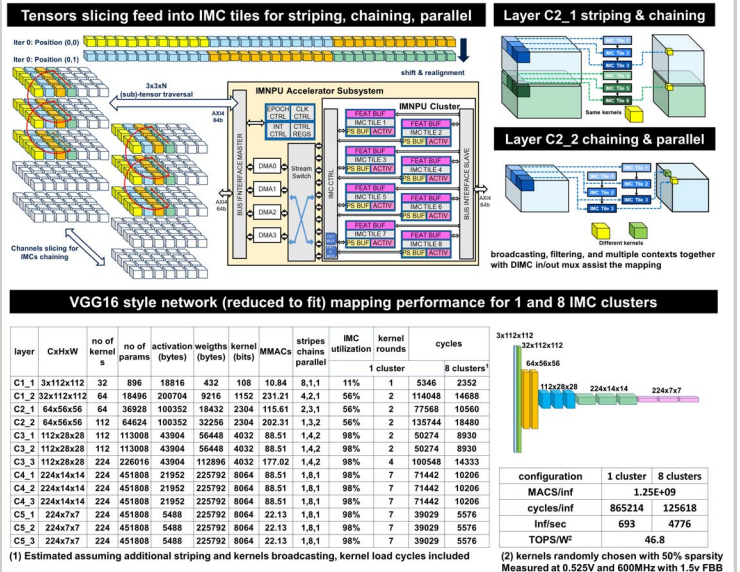
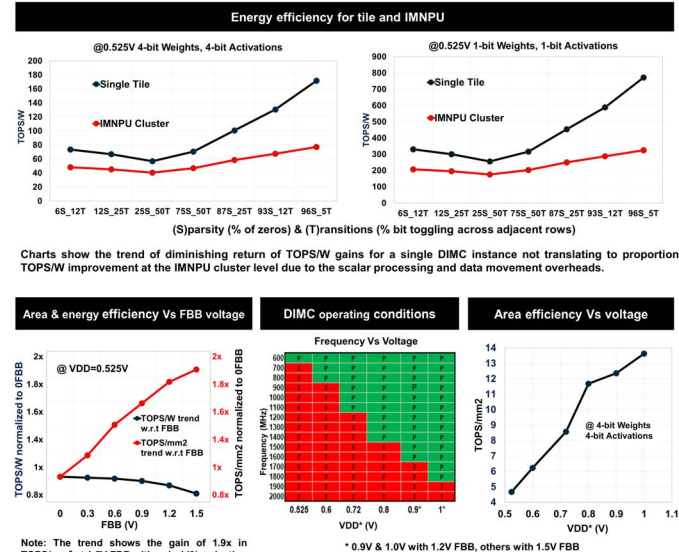
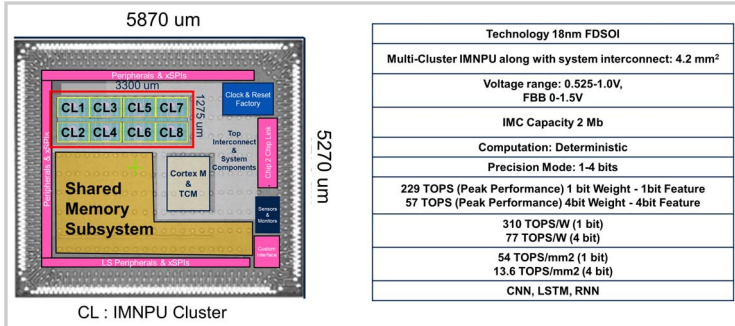


Figure 16.7.4: DMAs are programmed to traverse the geometry according to mapping parameters generated by a graph compiler.

Figure 16.7.5: Measured energy and area efficiency results for different applications, w.r.t. FBB voltage, V_{DD} voltage and DIMC operating conditions.

	Not in-memory computing				In-memory computing			
	Desol ISSCC'17	Bankman ISSCC'18	Jiao ISSCC'20	Fujiwara ISSCC'22	Chih ISSCC'21	Jia ISSCC 21	This Work	
Technology	28nm FDSOI	28nm	12nm	5nm	22nm	16nm	18nm FDSOI	
Area (mm ²)	34.8 mm ²	6	709	0.0133mm ²	0.202mm ²	25	4.2 mm ²	
Voltage (V)	0.575 - 1.1	0.6 0.8	0.85	0.5-0.9V	0.72V	0.8	0.525-1.0V	
On-chip Memory	41.9 Mb	2.6 Mb	1536 Mb	64 Kb ¹	64 Kb ¹	4.5 Mb ¹	DIMC Capacity 2 Mb ¹	
Deterministic or Approximate computation	Deterministic	Approximate	Deterministic	Deterministic	Deterministic	Approximate	Deterministic	
Bit Precision	8-16 b	1 b	8 b 16 b	4 b	4 b (weight)	1 - 8 b	1-4b	
Demonstrated System Scalability	YES	NO	YES	NO	NO	YES	YES	
Peak MAC Throughput (GOPS) ²	750	3.75 25	3.3M	2.8K	3300	11.8K ²	57K ²	
Peak MAC Energy Eff. (TOPS/W) ²	5.8	48.3 33.4	11.8	254 (Single Tile)	89	121 ²	77 (System 18FD)	
MAC Comp. Density (TOPS/mm ²) ²	2	0.0024 0.016	4.65	221 (Single Tile)	16.3	2.67 ²	13.6 (System 18 FD) 244 ² (System 5nm)	
Application	CNN	CNN	CNN	Matrix-Vect Multi.	Matrix-Vect. Multi.	CNN, RNN, MLP, etc.	CNN, LSTM, RNN	

¹ Only consider memory In-memory Computing ² Normalized to 4-bit compute ³ TOPS/mm² Scaled from 18FD to 5nm FF assuming an 18x combined scaling. ⁴ Number by exploring weight/input sparsity similar to [5]



Ultra-low-power always-on staged inference application example

Endurance for a battery-operated device for video surveillance, a single cluster is always on clocked @ 10 MHz running a VGG16-like network (as in Fig. 16.7.4) while the rest of the SoC is powered down and selectively activated @ 400 MHz to process a 10x complexity network (e.g., a ResNet152) with a duty cycle of 1 to 100 at 0.525 VDD

configuration	MACSS/inf	Inf/sec	IMNPU Power	Total Power ¹	Battery endurance ² (1/100 duty cycle)
1 cluster @ 10 MHz, 0.0V FBB	1.25 GOPS	10	267 uW	567 uW	363 days
8 clusters @ 400MHz, 0.3V FBB	12.5 GOPS	30	8.0 mW	12.0 mW	

(1) Estimated power includes a portion of shared memory, I/Os, clock, and external sensor interface
(2) 6000 mA/h battery capacity assumed (e.g., 2 AA 1.5v batteries)

Figure 16.7.7: Die photo and metrics of the multi-cluster IMNPU system prototype and always-on application example.